



AMRITA
VISHWA VIDYAPEETHAM

School of
Engineering

21ARE495 PROJECT PHASE - I

WEED REMOVAL Autonomous Mobile Robot

Guide:- Pramod sreedharan

Siddanth Bhogju-AM.EN.U4ARE22015

Pothula Adithya-AM.EN.U4ARE22026

Revu Danusith Kumar-AM.EN.U4AARE22049

INTRODUCTION & OBJECTIVES

INTRODUCTION

Precision agriculture demands tools that can intelligently differentiate crops from weeds and remove them effectively without affecting surrounding plants. Manual weeding remains labor-intensive and chemically inefficient. To address this, our project aims to develop a Dual-Mode Delta Robot Weeder AMR, an robotic system capable of performing both mechanical and chemical weed removal using a single high-speed parallel manipulator platform.

OBJECTIVES

1. To design a Delta-style 3-DoF manipulator capable of rapid and precise positioning.
2. To integrate a dual-mode end-effector combining a servo-electric gripper and spray nozzle.
3. To implement perception and control algorithms using ROS2, MoveIt2, and MPC.
4. To train and deploy a YOLOv8-based weed detection model for intelligent targeting.
5. To achieve field-ready simulation and testing through a unified robotic control stack

CAD SYSTEM OVERVIEW

SYSTEM STRUCTURE

The mechanical structure is based on a 3-DoF Delta manipulator, modeled in SolidWorks. Chosen for its high speed, stiffness, and positional accuracy.

Key Design Considerations:

- ✓ Weight reduction
- ✓ Structural stability
- ✓ Workspace optimization

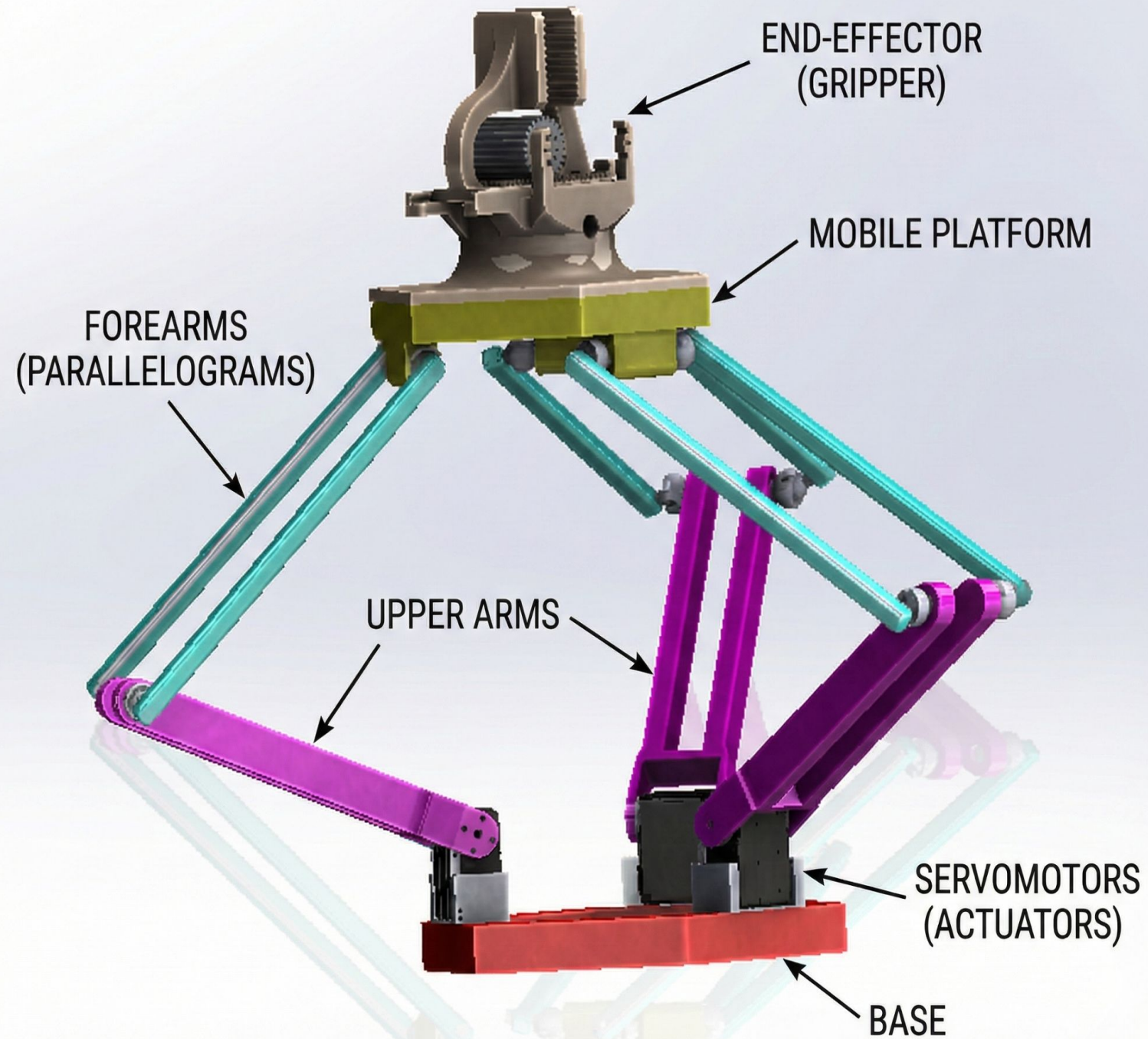
CORE COMPONENTS

- ✓ **Fixed Base Plate:** With motor mounts
- ✓ **Three Delta Arms:** Identical kinematic chains
- ✓ **Moving Platform:** End-effector carrier
- ✓ **Dual-Mode Head:** Integrated tool

ASSEMBLY VIEW

This assembly demonstrates the parallel structure responsible for precise vertical and planar motion.

- ✓ **Red Base:** Houses the servo motors
- ✓ **Purple Links:** Upper arms
- ✓ **Cyan Links:** Lower parallelogram arms
- ✓ **End-Effector:** Mounted at the front
- ✓ **Joints:** Ball-socket connections for flexibility



DELTA ARM STRUCTURE

Each arm of the delta robot consists of one upper rigid link connected to the base motor and two lower links forming a parallelogram and each arm is connected with a ball socket joint.

Mechanism Advantages

stiffness, low backlash, and improved positioning accuracy.

The parallel configuration maintains constant orientation while allowing smooth translation.



| JOINT CONFIGURATION



REVOLUTE JOINTS

Located at the base motors to provide the primary actuation force.



UNIVERSAL JOINTS

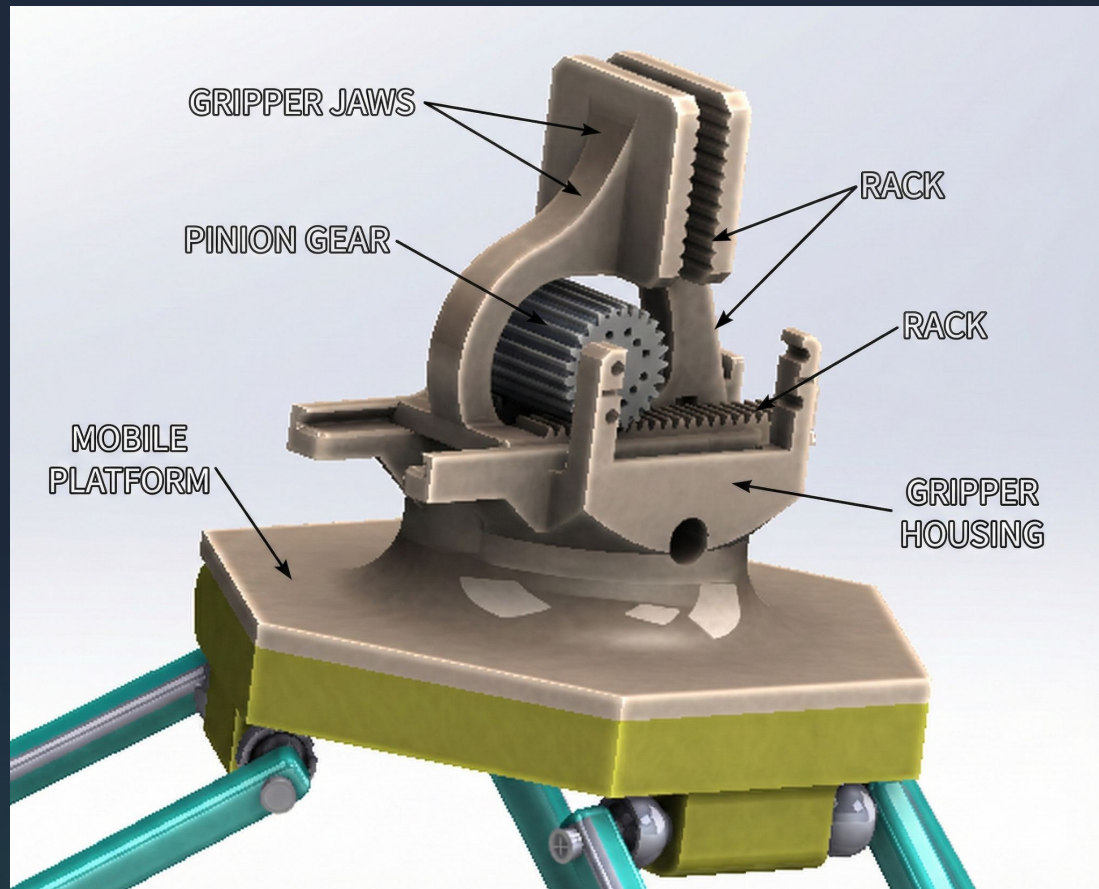
Used in mid-links to allow flexible movement while maintaining rigidity.



SPHERICAL JOINTS

At the end-effector to ensure smooth articulation and precise control.

END EFFECTOR DESIGN



FUNCTIONAL UNIT

A compact and lightweight module integrating two critical functions for weed management.

- ✓ **Mechanical:** Servo-controlled 2 finger gripper for physical weed pulling.
- ✓ **Chemical:** Micro spray nozzle for targeted herbicide application.

a tube needs to be inserted near the end effector

CAD DESIGN ADVANTAGES



MODULAR & STABLE

Modular design for easy manufacturing and maintenance.
Balanced load distribution prevents tipping.



OPTIMIZED PERFORMANCE

Reduced vibration due to optimized link geometry and
lightweight structure reduces motor strain.



OPERATIONAL EFFICIENCY

Large operational workspace suitable for varied field conditions, ensuring better system longevity.

VISION SYSTEM OVERVIEW

detection of weeds using camera input and OpenCV-based processing. (YOLOV8n)

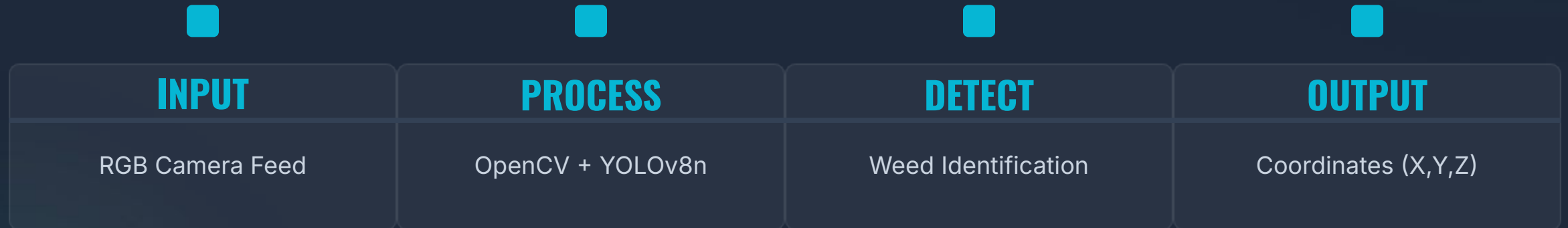


IMAGE PROCESSING FLOW

PREPROCESSING

- ✓ Capturing live frames from RGB Camera
- ✓ Resizing for model input
- ✓ Normalization of pixel values

DETECTION & OUTPUT

- ✓ Detection via YOLOv8 model
- ✓ Generation of Bounding Boxes
- ✓ Confidence Scores Calculation
- ✓ Weed Class Labeling

YOLOV8 MODEL TRAINING

The learning model is trained using a agricultural dataset to ensure robust performance under varying lighting and backgrounds.

640px

IMAGE SIZE

75

EPOCHS

16

BATCH SIZE



| FROM 2D TO 3D LOCALIZATION



2D DETECTION

Pixel coordinates (u, v) extracted from the camera frame using OpenCV.



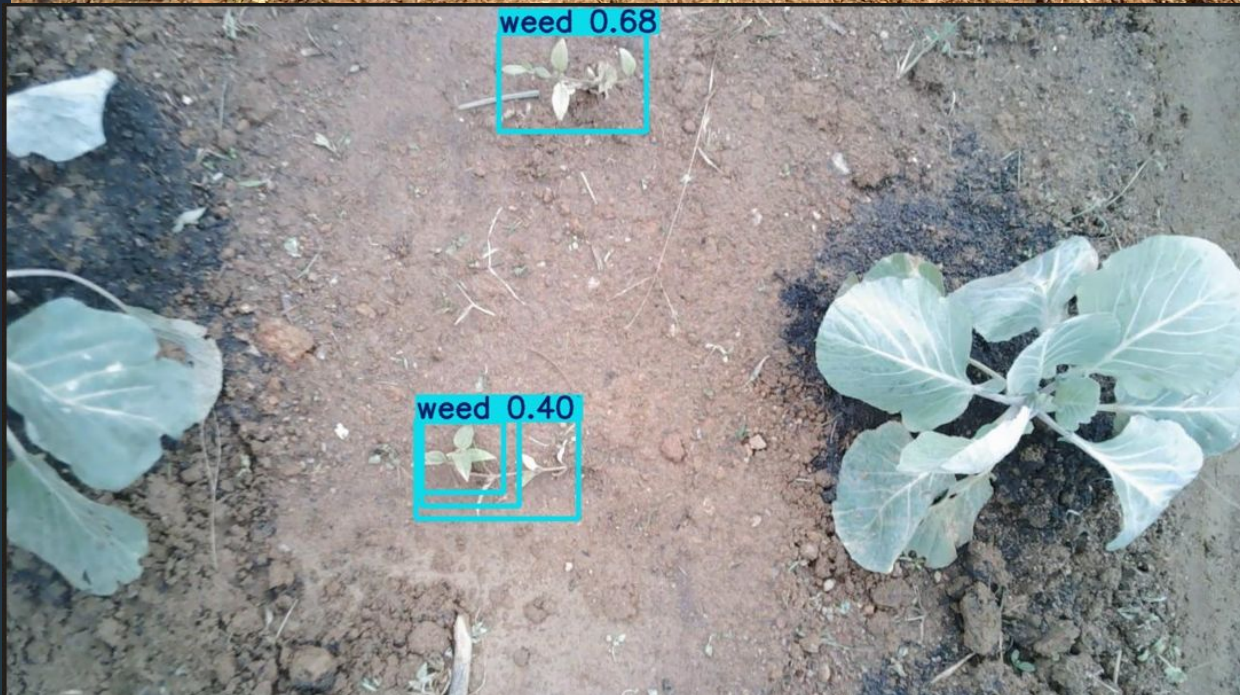
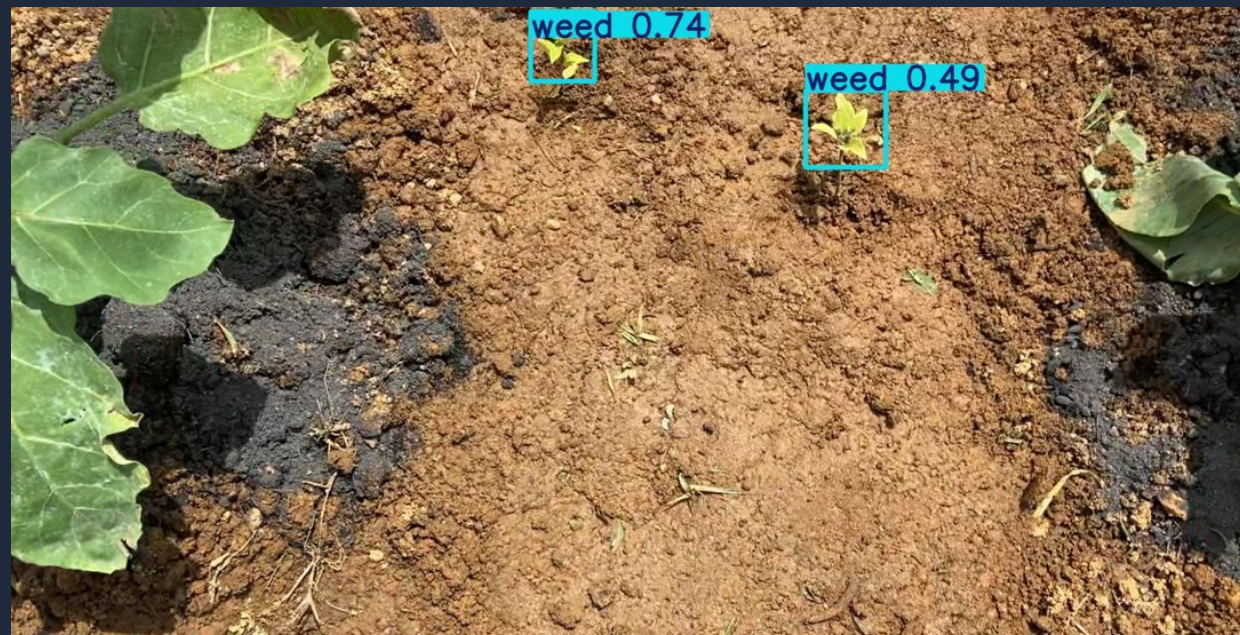
DEPTH ESTIMATION

Distance (z) obtained using stereo camera data.



TRANSFORMATION

Coordinate transformation using ROS2 tf2 to map to Real-World (X, Y, Z) .



VISION OUTPUT

This output displays bounding boxes around detected weeds, along with confidence values.

The detected region is used as the target point for the robot arm.

Validation: This visual feedback confirms detection accuracy and validates system performance.

DELTA ROBOT KINEMATICS

Kinematics governs the motion of the robot based on joint configurations.

INVERSE KINEMATICS

Target → Angles

FORWARD KINEMATICS

Angles → Position

INVERSE KINEMATICS (IK)

Given a target position $P(x, y, z)$, calculate the required joint angles $\theta_1, \theta_2, \theta_3$.

HTM derivation of elbow joint

Target weed position in Cartesian coordinates:

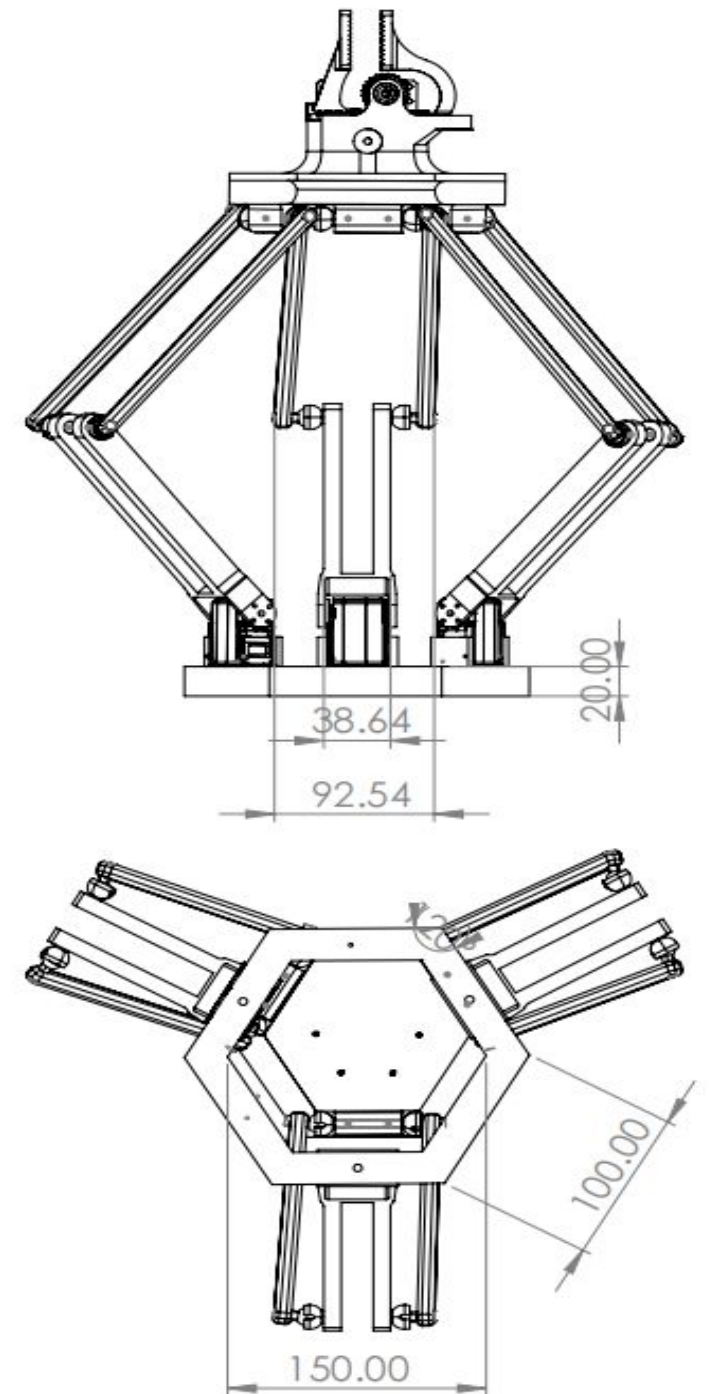
$${}^{Base}T_{J_i} = T_{RotZ}(\phi_i) \cdot T_{TransX}(R) \cdot T_{RotY}(\theta_i) \cdot T_{TransX}(L)$$

Where ϕ_i is the azimuthal angle ($0^\circ, 120^\circ, 240^\circ$) and θ_i is the active motor angle. Multiplying these matrices yields the elbow coordinates vector J_i :

$$J_i = \begin{bmatrix} (R + L \cos \theta_i) \cos \phi_i \\ (R + L \cos \theta_i) \sin \phi_i \\ -L \sin \theta_i \end{bmatrix}$$

We solve for θ_i using the tangent half-angle substitution:

$$\|J_i - E_i\|^2 = l^2 \quad \theta_i = 2 \arctan \left(\frac{-F_i \pm \sqrt{E_i^2 + F_i^2 - G_i^2}}{G_i - E_i} \right)$$



FORWARD KINEMATICS (FK)

Given joint angles $\theta_1, \theta_2, \theta_3$, calculate the end-effector position $P(x, y, z)$.

Since the elbow positions J_1, J_2, J_3 are known (derived from θ_i), the problem is geometrically equivalent to finding the intersection point of three spheres, each centered at a "virtual elbow" J'_i with radius l .

1. **Shift Centers:** We virtually shift the elbow centers inward by the platform radius r . Let the virtual centers be C_i :

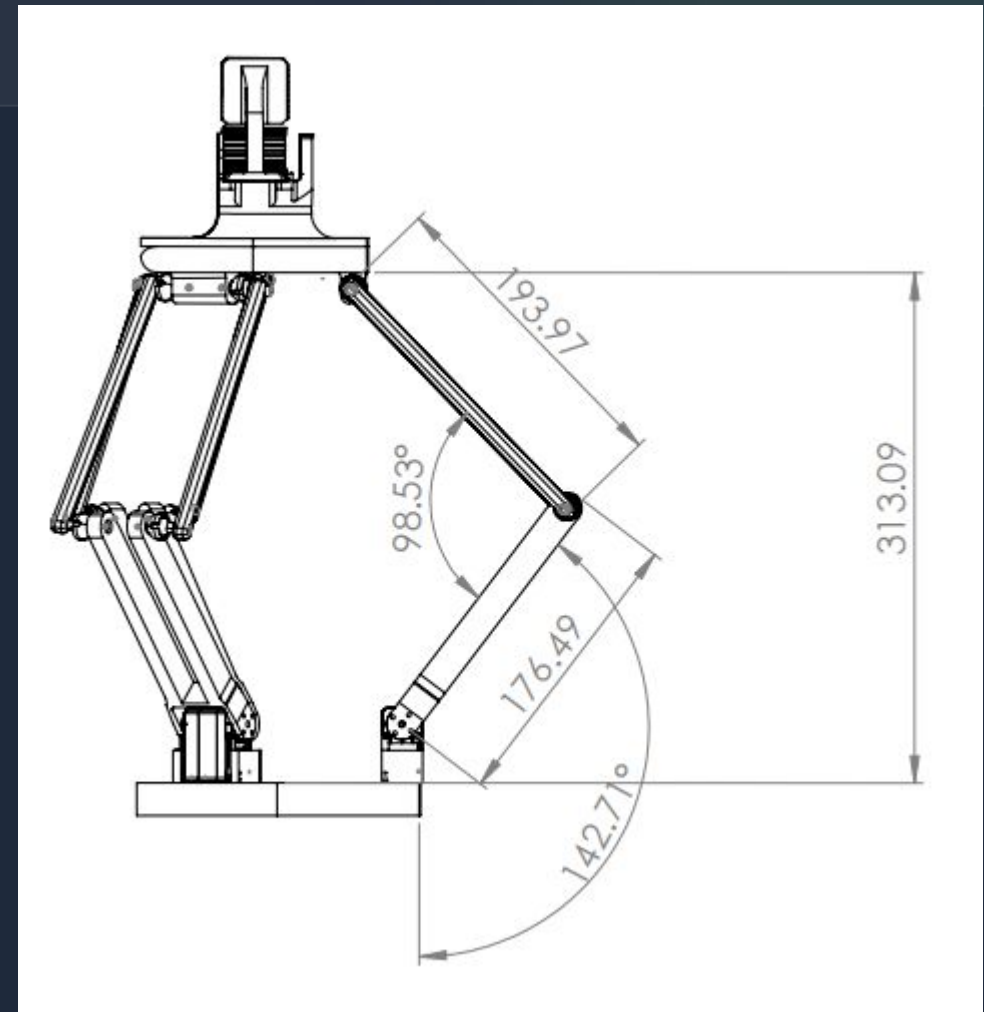
$$C_i = \begin{bmatrix} J_{ix} - r \cos \phi_i \\ J_{iy} - r \sin \phi_i \\ J_{iz} \end{bmatrix}$$

2. **Sphere Equations:** The end-effector $P(x, y, z)$ must satisfy:

$$(x - C_{ix})^2 + (y - C_{iy})^2 + (z - C_{iz})^2 = l^2 \quad \text{for } i = 1, 2, 3$$

3. **Algorithmic Solution:**

- Subtracting the sphere equations eliminates the quadratic terms (x^2, y^2, z^2) and yields linear equations for planes.
- The intersection of these planes defines a line.
- The intersection of this line with one of the spheres yields two points. We select the point with the larger Z value (for a downward-pointing configuration).



Workspace analysis

Based on the parameters ($L = 176.49$, $l = 193.97$), the workspace geometry is a "rounded hexagonal paraboloid."

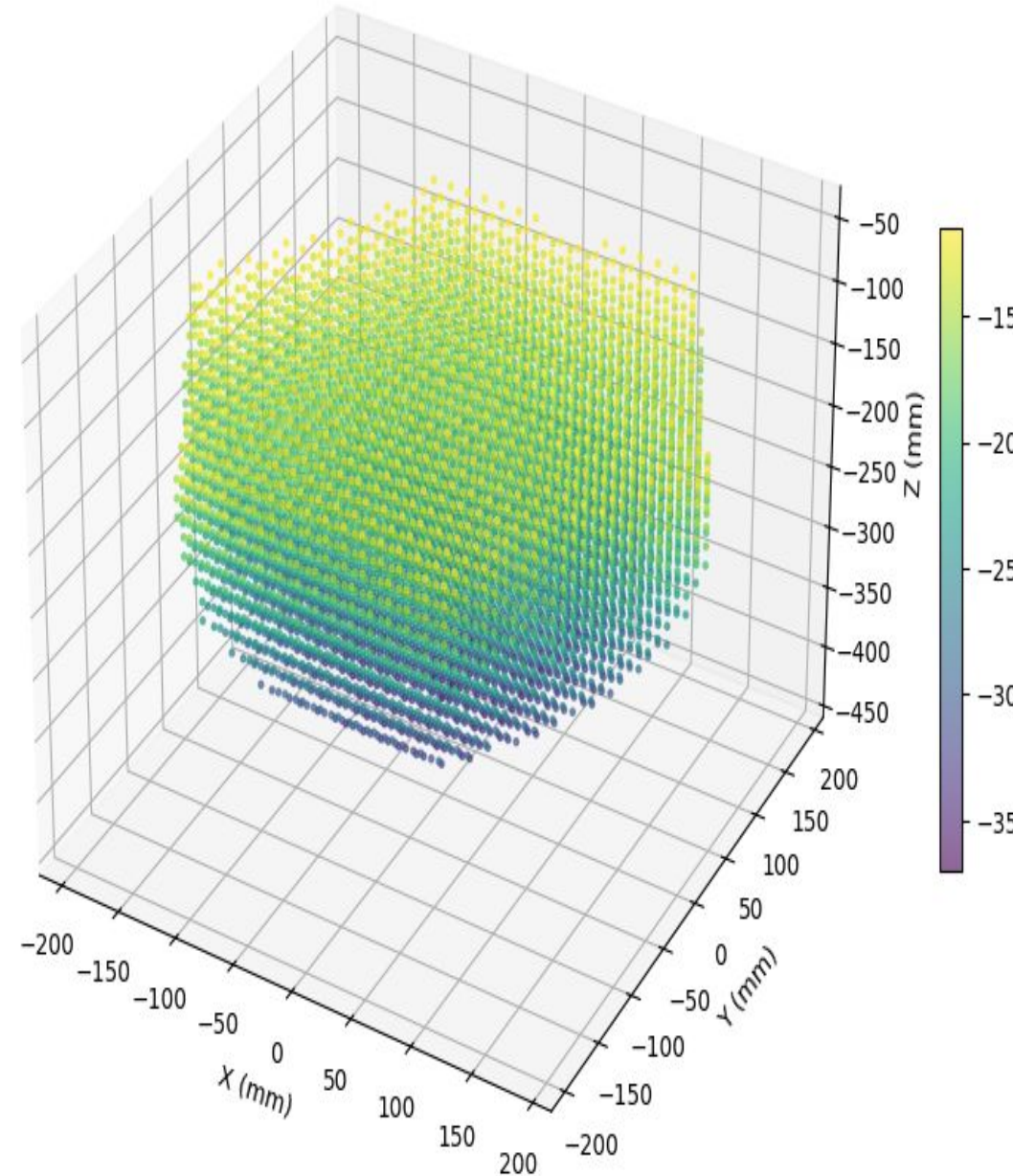
- Maximum Vertical Reach ($Z_{\max}$): Occurs when the active and passive arms are fully extended vertically.

$$Z_{\max} \approx L + l = 176.49 + 193.97 = 370.46 \text{ mm}$$

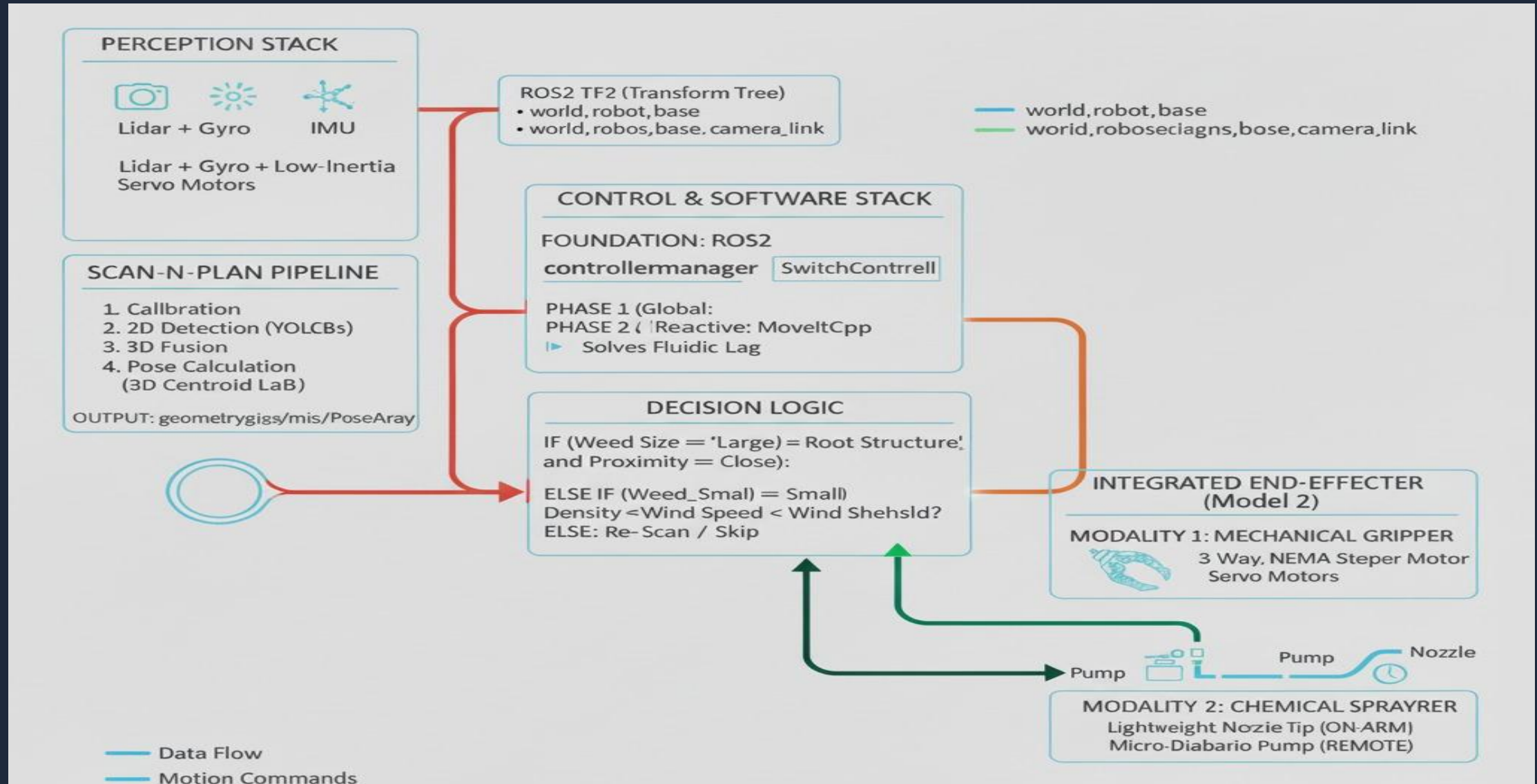
- Operational Radius (R_{work}): At the nominal operating depth (approx. $Z = 240$ mm), the robot covers a circular area sufficient for weed targeting.

$$R_{\text{work}} \approx 135 \text{ mm}$$

Calculated Workspace for $L=176.49$, $l=193.97$



CONTROL WORKFLOW



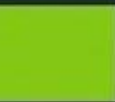
| CONCLUSION & FUTURE WORK

Conclusion

- ✓ CAD design is finalized
- ✓ weed detection implemented
- ✓ kinematic control validated
- ✓ Validated workspace

FUTURE WORK

- ✓ Hardware testing
- ✓ Full autonomous navigation
- ✓ model enhancement (More species)
- ✓ Real-time hardware integration



THANK YOU